

Contact	Department of Biology Tufts University 200 College St Medford, MA 02155 Lawrence.Uricchio@tufts.edu	https://scholar.google.com/citations?user=XyVUpZYAAAAJ uricchio.github.io
Positions	2021- <i>Assistant Professor</i> , Department of Biology, Tufts University	
Education	2018-2021 <i>Postdoctoral Scholar</i> , Integrative Biology, University of California, Berkeley Advisors: Dr. Mike Boots & Dr. Becca Tarvin 2015-2018 <i>Postdoctoral Fellow</i> , Biology, Stanford University Advisors: Dr. Noah Rosenberg & Dr. Erin Mordecai 2014 <i>PhD</i> , Bioinformatics, University of California, San Francisco PhD advisor: Dr. Ryan Hernandez 2011 <i>MS</i> , Computer Science, University of Chicago 2009 <i>MS</i> , Biophysical Sciences, University of Chicago 2005 <i>BA</i> , Physics, Carleton College	
Honors	2021- Youniss Family Assistant Professor of Innovation 2016-2018 NIH IRACDA Postdoctoral Fellow, Stanford University & SJSU 2015-2016 CEHG Postdoctoral Fellow, Stanford University 2014 Discovery Fellow, UCSF 2014 ASHG trainee award for excellence in human genetics research, semi-finalist 2013 Teaching Assistant Excellence Award, UCSF 2012-2014 Achievement Rewards for College Scientists Fellow, UCSF 2005 Distinction in the physics major, Carleton College 2005 Phi Beta Kappa (academic honor society), Carleton College 2005 Sigma Xi (scientific research honor society), Carleton College 2005 Magna Cum Laude, Carleton College 2002 Dean's list, Carleton College	

Articles & conference papers

†: *corresponding author*; *: *equal contributions*

In progress

- [1] WRIGHT-LICHTER JX, LEAV A, GORDON S, **URICCHIO LH**[†] (2025) Multimodal selection and the evolution of plumage on a warming planet. **submitted**.
- [2] DABHOLKAR S, GOUVEA J, COBURN K, **URICCHIO LH**. Using “Resistance” Spotlights to Support Studentsâ Critical Reflections on and Reimaginings of Disciplinary Values and Participation. **in revision**.
- [3] LELE K[†], **URICCHIO LH**[†] (2026) Fitness landscapes for species interactions: when do population genetics and adaptive dynamics diverge?. *bioRxiv*, **submitted**.
- [4] CALDERON A, KOZAK GM, URICCHIO LH[†] (2026) Leveraging the joint site frequency spectrum to detect genomic regions of early divergence. *bioRxiv*, **submitted**.

Peer reviewed

- [1] LELE K, WOLFE BE, **URICCHIO LH**[†] (2026) Pairwise interactions and serial bottlenecks help explain species composition in a multispecies microbial community. *Ecology*, **107**(1), e70284.
- [2] WUITCHIK DM, FIFER JE, HUZAR AK, PECHENICK JA, **URICCHIO L**, DAVIES SW (2025) Outside your shell: exploring genetic variation in two congeneric marine snails across a latitude and temperature gradient. *BMC Genomics*, **26**, 486.

- [3] GEARTY W, **URICCHIO LH**, LYONS K (2024) Investigating the Biotic and Abiotic Drivers of Body Size Disparity in Communities of Non-Volant Terrestrial Mammals. *Global Ecology and Biogeography*, p. e13913.
- [4] MURGA-MORENO J, CASILLAS S, BARBADILLA A, **URICCHIO L[†]**, ENARD D[†] (2024) An efficient and robust ABC approach to infer the rate and strength of adaptation. *G3: Genes, Genomes, Genetics*, **14**(4), jkae031.
- [5] LOUW N, WOLFE BE, **URICCHIO LH[†]**. (2024) A phylogenomic perspective on interspecific competition. *Ecology Letters*, **27**(2), e14359.
- [6] DABHOLKAR S, GOUVEA J, **URICCHIO LH**. (2023) Designing for and Characterizing Critical Navigation of Disciplinary Values in an Undergraduate Computational Biology Course. In *Proceedings of the 17th International Conference of the Learning Sciences-ICLS 2023*, pp. 1386-1389 International Society of the Learning Sciences.
- [7] WAN YC, NAVARRETE MÉNDEZ M, O'CONNELL LA, AND **URICCHIO LH**, AND ROLAND AB, MAAN ME RON SR, BETANCOURTH-CUNDAR M et al. (2023) Selection on visual opsin genes in diurnal Neotropical frogs and loss of the SWS2 opsin in poison frogs. *Molecular Biology and Evolution*, **40**(10), msad206.
- [8] **URICCHIO LH**, BRUNS EL, HOOD M, BOOTS M, ANTONOVICS J. (2023) Multimodal pathogen transmission as a limiting factor in host distribution. *Ecology*, **104**(3), e3956.
- [9] VISHER E, **URICCHIO L**, BARTLETT L, DENAMUR N, YARCAN A, ALHASSANI D, BOOTS M. (2022) The evolution of host specialization in an insect pathogen. *Evolution*, **76**(10), 2375–2388.
- [10] EBEL ER, **URICCHIO LH**, PETROV DA, EGAN ES. (2022) Revisiting the malaria hypothesis: accounting for polygenicity and pleiotropy. *Trends in Parasitology*, **38**(4).
- [11] COUPER LI, FARNER JE, CALDWELL JM, CHILDS ML, HARRIS MJ, KIRK DG, NOVA N, SHOCKET M, SKINNER EB, **URICCHIO LH**, EXPOSITO-ALONSO M, MORDECAI EA. (2021) How will mosquitoes adapt to climate warming?. *eLife*, **10**, e69630.
- [12] SONG C, **URICCHIO LH**, MORDECAI EA, SAAVEDRA S. (2021) Understanding the emergence of contingent and deterministic exclusion in multispecies communities. *Ecology Letters*, **24**, 2155â2168.
- [13] **URICCHIO LH[†]**. (2020) Evolutionary perspectives on polygenic selection, missing heritability, and GWAS. *Human Genetics*, **139**(1), 5–21.
- [14] SEVERSON AL, **URICCHIO LH**, ARBISSER IM, GLASSBERG EC, ROSENBERG NA. (2019) Analysis of author gender in TPB, 1991-2018. *Theoretical Population Biology*, **127**, 1–6.
- [15] **URICCHIO LH**, PETROV DA, ENARD D. (2019) Exploiting selection at linked sites to infer the rate and strength of adaptation. *Nature Ecology & Evolution*, **3**(6), 977–984.
- [16] HERNANDEZ RD, **URICCHIO LH**, HARTMAN K, YE C, DAHL A, ZAITLEN N. (2019) Ultrarare variants drive substantial cis heritability of human gene expression. *Nature Genetics*, **51**(9), 1349–1355.
- [17] **URICCHIO LH***, KITANO HC*, GUSEV A, ZAITLEN NA. (2019) An evolutionary compass for detecting signals of polygenic selection and mutational bias. *Evolution Letters*, **3**(1), 69–79.
- [18] **URICCHIO LH[†]**, DAWS SC, SPEAR ER, MORDECAI EA[†]. (2019) Priority effects and nonhierarchical competition shape species composition in a complex grassland community. *The American Naturalist*, **193**(2), 213–226.
- [19] GIGNOUX CR, TORGERSON DG, PINO-YANES M, **URICCHIO LH**, GALANTER J, ROTH LA, ENG C, HU D, NGUYEN EA, HUNTSMAN S et al. (2019) An admixture mapping meta-analysis implicates genetic variation at 18q21 with asthma susceptibility in Latinos. *Journal of Allergy and Clinical Immunology*, **143**(3), 957–969.
- [20] GOLDBERG A, **URICCHIO LH**, ROSENBERG NA (2018) Natural selection in human populations. *Oxford Bibliographies in Evolutionary Biology*, DOI:10.1093/OBO/9780199941728-0112.
- [21] **URICCHIO LH**, WARNOW T, ROSENBERG NA (2016) An analytical upper bound on the number of loci required for all splits of a species tree to appear in a set of gene trees. *BMC Bioinformatics*, **17**(14), 241–250.
- [22] **URICCHIO LH[†]**, ZAITLEN NA, YE CJ, WITTE JS, HERNANDEZ RD[†]. (2016) Selection and explosive growth alter genetic architecture and hamper the detection of causal rare variants. *Genome Research*, **26**, 863–873.

- [23] **URICCHIO LH**, TORRES R, WITTE JS, HERNANDEZ RD. (2015) Population genetic simulations of complex phenotypes with implications for rare variant association tests. *Genetic Epidemiology*, **39**(1), 35–44.
- [24] **URICCHIO LH**, HERNANDEZ, RD. (2014) Robust forward simulations of recurrent hitchhiking. *Genetics*, **197**(1), 221–236.
- [25] MAHER MC*, **URICCHIO LH***, TORGERSON DG, HERNANDEZ RD. (2013) Population genetics of rare variants and complex diseases. *Human Heredity*, **74**(3-4), 118–128.
- [26] **URICCHIO LH**, CHONG JX, ROSS KD, OBER C, NICOLAE DL. (2012) Accurate imputation of rare and common variants in a founder population from a small number of sequenced individuals. *Genetic Epidemiology*, **36**(4), 312–319.
- [27] ÇALIŞKAN M, CHONG JX, **URICCHIO L**, ANDERSON R, CHEN P *et al.* (2011) Exome sequencing reveals a novel mutation for autosomal recessive nonsyndromic mental retardation in the *TECR* gene on chromosome 19p13. *Human Molecular Genetics*, **20**(7), 1285–1289.
- [28] MILLER LS, PIETRAS EM, **URICCHIO LH**, HIRANO K, RAO S *et al.* (2007) Inflammasome-mediated production of IL-1 β is required for neutrophil recruitment against *Staphylococcus aureus* in vivo. *Journal of Immunology*, **179**(10), 6933–6942.
- [29] PATTANAYAK AK, BROOKS DWC, DE LA FUENTE A, **URICCHIO L**, HOLBY E *et al.* (2005) Coarse-grained entropy decrease and phase-space focusing in Hamiltonian dynamics. *Physical Review A*, **72**(1), 013406.

Dissertation

- [1] **URICCHIO LH**. (2014) Models and forward simulations of selection, human demography, and complex traits. UNIVERSITY OF CALIFORNIA, SAN FRANCISCO.

Teaching	2026	Instructor, <i>Computational Laboratory in Population Genomics</i> , Bio 145, Tufts University
	2025	Instructor, <i>Computational Biology</i> , Bio 35, Tufts University
	2025	Instructor, <i>Computational Laboratory in Population Genomics</i> , Bio 145, Tufts University
	2024	Instructor, <i>Computational Biology</i> , Bio 35, Tufts University
	2022	Guest lecture, <i>R and Programming for STEM</i> , Bio 196, Tufts University
	2022	Instructor, <i>Biostatistics</i> , Bio 132, Tufts University
	2022	Instructor, <i>Computational Biology</i> , Bio 35, Tufts University
	2022	Instructor, <i>Seminar in Ecology and Evolution</i> , Bio 244, Tufts University
	2022	Guest lecture, <i>Resilience seminar</i> , Bio 263, Tufts University
	2021	Instructor, <i>Computational Biology</i> , Bio 35, Tufts University
	2018	Co-instructor, <i>Evolutionary genetics</i> , San Jose State University
	2017	Co-instructor, <i>Ecology</i> , San Jose State University
	2017	Co-founder, <i>Stanford postdoc pedagogy miniseries</i> , Stanford University
	2016-2017	Guest lecturer, <i>Evolution</i> , Stanford University
	2014	Graduate teaching assistant, <i>Computational Evolutionary Genomics</i> , UCSF
	2013	Graduate student instructor, <i>Computational Biology</i> , UC Berkeley
	2009	Guest lecturer, <i>Genes, Networks, and Cells</i> , University of Chicago
	2003-2004	Teaching assistant, tutor, & grader, <i>Introduction to Physics, Classical & Computational Mechanics, Contemporary Experimental Physics</i> , Carleton College

Mentorship	2024-	Undergrad research mentor for Stephen Burchfield, Tufts Biology Dept
	2024-	Postdoc mentor for Dr. Daniel Wuitchik, Tufts Biology Dept
	2023-	Graduate co-mentor for Kaitlyn Coburn, Tufts Biology Dept
	2022-2024	Undergrad research mentor for Alex Hovsepian, Tufts Biology Dept & Summer Fellow
	2022-2024	Undergrad research mentor for Daphne Garcia, Tufts Biology Dept & Summer Fellow
	2022	Undergrad research mentor for Alison Leav, Tufts Biology Dept
	2021-2022	Postdoc mentor for Dr. Adam Pepi, Tufts Biology Dept
	2021-	PhD mentor for Kasturi Lele, Tufts Biology Dept
	2021-	PhD mentor for Alejandro Calderon, Tufts Biology Dept
	2021-2023	PhD mentor for SJ McGeady, Tufts Biology Dept (graduated with MS)
	2021	Summer research mentor for Alice Lau, Tufts MS student and TIE Environmental Fellow
	2021	Summer research mentor for SJ McGeady, incoming Tufts PhD student
	2016	Summer research co-mentor for Alan Aw, Stanford undergraduate
	2013	Summer research co-mentor for Isela Hernandez, Biological Health Sciences Internship Program
	2010	Summer research co-mentor for Sam Neal, Summer Link High School Program
Service	2026	NSF Panelist (DEB)
	2025-2026	Tufts Biology Department Graduate Admissions Committee
	2025-2026	Tufts University Faculty DEIJ Committee
	2024-2025	Tufts University Faculty Committee on Work/Life
	2024	NSF Panelist (DEB)
	2023	Tufts Biology DEIJ Co-chair, spring semester
	2023	NSF <i>ad hoc</i> grant reviewer
	2022	Canada Foundation for Innovation Grant Reviewer
	2021	NSF panelist (IOS)
	2022	Committee Member, lecturer search committee, Tufts Biology Department
	2021-2025	Committee Member, Tufts Biology DEIJ committee (on leave for 2023-2024)
	2013-	Invited peer reviewer for <i>Genetics</i> , <i>BMC Evolutionary Biology</i> , <i>PLoS Genetics</i> , <i>Molecular Biology & Evolution</i> , <i>Genome Biology & Evolution</i> , <i>Molecular Ecology</i> , <i>Heredity</i> , <i>Theoretical Population Biology</i> , <i>Nature Genetics</i> , <i>American Journal of Human Genetics</i> , <i>Entropy</i> , <i>Functional Ecology</i> , <i>Peer Community in Evolutionary Biology</i> , <i>Journal of Allergy and Clinical Infection</i> , <i>eLife</i> , <i>PLoS ONE</i> , <i>G3</i> , <i>Evolution</i> , <i>IEEE/ACM Transactions on Computational Biology and Bioinformatics</i> , <i>EMPH: Evolution, Medicine & Public Health</i> , <i>Ecological Modelling</i> , <i>Human Population Genetics and Genomics</i>
	2016	Committee Member & Presenter, Stanford Postdoc Pedagogy Journal Club
	2015-2016	Committee Member, Stanford CEHG diversity outreach committee
Competitive Funding	2022	Tufts Springboard Award, <i>Disciplinary Identity Development in Computational Biology</i> (\$15,000)
	2016-2018	NIH IRACDA Fellow, Stanford & SJSU (\$53,600 per year)
	2016-2017	Stanford Teaching & Mentoring Academy Award (\$6,870)
	2015-2016	Stanford CEHG Postdoctoral Fellowship (\$50,000)
	2012-2014	UCSF Achievement Rewards for College Scientists Fellow (\$12,000 per year)
	2014	UCSF Discovery Fellow (\$4,000 for travel & research over 2 years)
News coverage	2026	Lawrence interviewed on People Behind the Science
	2026	<i>Tufts Now</i> on microbial community assembly
	2019	<i>Stanford News</i> on priority effects in grasslands

Selected presentations

- [1] Interrogating the Role of Rare Alleles in Resilience to Environmental Change. *Organismal Resilience Meeting of the Linnean Society, London, England*. **Talk** (2025).
- [2] The role of mutation spectra in determining ecological outcomes. *Departmental Seminar, UMass Dartmouth, Dartmouth*,

MA. **Seminar** (2025).

- [3] Multimodal selection and the evolution of thermal performance on a warming planet. *American Society of Naturalists Meeting, Monterey, CA.* **Talk** (2025).
- [4] Integrating ecological processes into evolutionary models & inference. *UNCC Bioinformatics Departmental Seminar, Charlotte, NC.* **Seminar** (2024).
- [5] The influential role of physics-based methods in evolutionary biology. *Carleton College, Northfield, MN.* **Seminar**, 2024.
- [6] What can we learn by integrating ecological processes into evolutionary inference?. *Cornell University, Ithaca, NY.* **Seminar**, 2024.
- [7] Eco-evolutionary perspectives on microbial interactions, community assembly, and adaptation. *UMass Chan Medical School, Department of Microbiology and Physiological Systems, Worcester, MA.* **Seminar**, 2023.
- [8] Youniss Family Assistant Professorship in Innovation: Evolutionary genetics as a prospective discipline. *Tufts University, Medford, MA.* **Seminar**, 2023.
- [9] Evolutionary genetics of costly adaptation to pathogens. *American Naturalist, Asilomar, CA.* **Talk**, 2023.
- [10] Infectious disease and costly resistance in evolutionary genetics. *Tufts Biology Departmental Retreat, Medford, MA.* **Talk** (2022).
- [11] Back to the future: the population genetics of adaptation in our rapidly changing world. *CSU East Bay, Hayward, CA.* **Seminar**, 2020.
- [12] Adaptation and constraint through the lens of genome-wide association studies. *University of Wisconsin, Madison, WI.* **Seminar**, 2020.
- [13] Adaptation and constraint through the lens of genome-wide association studies. *Temple University, Philadelphia, PA.* **Seminar**, 2020.
- [14] Adaptation and constraint through the lens of genome-wide association studies. *Tufts University, Somerville, MA.* **Seminar**, 2020.
- [15] Adaptation and constraint through the lens of genome-wide association studies. *Vanderbilt University, Nashville, TN.* **Seminar**, 2020.
- [16] Evolutionary perspectives on human complex traits and missing heritability. *CSUN, Northridge, CA.* **Seminar**, 2019.
- [17] Can modern genomic sequence data fulfill their evolutionary potential?. *Indiana University, Bloomington, IN.* **Seminar**, 2019.
- [18] Statistical inference of evolutionary processes from genomic data. *Macalester College, St. Paul, MN.* **Seminar**, 2019.
- [19] Can modern genomic sequence data fulfill their evolutionary potential?. *Cornell University, Ithaca, NY.* **Seminar**, 2019.
- [20] Exploiting selection at linked sites to infer the rate and strength of adaptation. *Evolution Meeting, Providence, RI.* **Talk**, 2019.
- [21] Genome-scale inference of adaptive evolution: new approaches for answering old questions. *Boise State University, Boise, ID.* **Seminar**, 2019.
- [22] Genome-scale inference of adaptive evolution: new approaches for answering old questions. *Chapman University, Orange, CA.* **Seminar**, 2019.
- [23] Evolutionary processes shaping the human genome. *Linfield College, McMinnville, OR.* **Seminar**, 2018.
- [24] Scientific teaching workshops for Stanford postdocs. *Stanford Education Day, Stanford, CA.* **Talk**, 2018.
- [25] Modulation of adaptation rate by background selection in the human genome. *Bay Area Population Genomics Meeting,*

Santa Cruz, CA. Talk, 2018.

- [26] Modulation of adaptation rate by background selection in the human genome. *American Society of Naturalists Meeting, Monterey, CA. Talk* (2018).
- [27] Designing and implementing scientific teaching workshops for postdocs. *Stanford Teaching and Mentoring Academy seminar series, Stanford, CA. Talk* (2018).
- [28] An analytical upper bound on the number of loci required for all splits of a species tree to appear in a set of gene trees. *RECOMB Comparative Genomics Meeting, Montreal, Canada. Talk* (2016).
- [29] Explosive growth and the genetic architecture of polygenic traits under selection. *American Society of Naturalists Meeting, Monterey, CA. Talk* (2016).
- [30] Selection and explosive growth may hamper the performance of rare variant association tests. *Bay Area Population Genomics Meeting, Stanford, CA. Talk* (2015).
- [31] Recent demography and natural selection hamper the power of rare variant association tests. *American Society of Human Genetics Meeting, San Diego, CA. Talk* (2014).
- [32] Model-based simulations of selection and demography with implications for heritable phenotypes and rare variant association tests. *UC Berkeley Center for Theoretical Evolutionary Genomics, Berkeley, CA. Talk* (2014).
- [33] Simulations and inference of simultaneous positive and negative selection. *Society of Molecular Biology and Evolution Meeting, San Juan, Puerto Rico. Poster* (2014).
- [34] Parameter rescaling for forward simulations of recurrent hitchhiking. *Bay Area Population Genomics Meeting, San Francisco, CA. Poster* (2013).
- [35] Forward simulations of recurrent selection and demographics with rescaled parameters. *American Society of Human Genetics Meeting, Boston, MA. Poster* (2013).
- [36] Accurate pedigree-based imputation. *Department of Human Genetics Seminar, University of Chicago, Chicago, IL. Talk* (2011).
- [37] Simultaneous measurement of myosin-II kinetics and cellular traction forces. *HHMI Interfaces Scholars Meeting, Chevy Chase, MD. Poster* (2008).